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Postflight Thermal Protection System Analysis of Hayabusa Reentry Capsule

Toshiyuki Suzuki* and Kazuhisa Fujita†
Japan Aerospace Exploration Agency, Tokyo 182-0012, Japan
and
Tetsuya Yamada,‡ Yoshifumi Inatani,§ and Nobuaki Ishii§
Japan Aerospace Exploration Agency, Kanagawa 252-0222, Japan

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The thermal response of the ablative thermal protection system of the Hayabusa capsule is calculated along the most probable reentry trajectory by using the integrated analysis method. The analysis method consists of an axisymmetric nonequilibrium Navier–Stokes flow solver and a radiative heat transfer analysis code loosely coupled to an ablation thermal response code. The calculated results are compared with the flight data obtained at the present time. It is found that a reasonable agreement is obtained between the calculated surface temperature and the temperature values estimated from the DC-8 airborne observation and the ground observation. In contrast, the calculated surface recession overestimates the measurement along the wall surface by a factor of three. It is also found that the density distributions inside the ablator can be reproduced reasonably both at the stagnation point and at the downstream region by taking account of the conventional surface reaction set and nitridation reaction with a laminar boundary condition. Especially for the thickness of the char layer at the downstream region, calculation overestimates the flight data when the effect of turbulence due to the pyrolysis gas is considered, whereas it underestimates the flight data when the effect of surface roughness of the ablative thermal protection system is taken into account.

Nomenclature

c_s	=	mass fraction of species s
J_s	=	mass flux of species s, $\mathrm{kg/(m^2 \cdot s)}$
k	=	surface reaction velocity, m/s
M_s	=	molecular weight of species s, kg/mol
q	=	heat flux, $\mathrm{W/m^2}$
R	=	universal gas constant, 8.314 $\mathrm{J/(mol \cdot K)}$
r	=	recession rate, m/s
S	=	surface recession, m
s	=	distance along wall surface, m
T	=	temperature, K
α	=	reaction probability
ε	=	emissivity
ρ	=	density, $\mathrm{kg/m^3}$
σ	=	Stefan–Boltzmann constant, $\mathrm{W/(m^2 \cdot K^4)}$

Subscripts

c	=	char
h	=	enthalpy transport due to species diffusion
nit	=	nitridation reaction
oxi	=	oxidation reaction
r	=	radiation
rw	=	radiation from wall surface
sub	=	sublimation reaction
t	=	translational temperature
v	=	virgin or vibration

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*Associate Senior Researcher, Aerospace Research and Development Directorate, 7-44-1 Jindaiji-Higashi-machi, Chofu-shi; suzuki.toshiyuki@jaxa.jp. Member AIAA.

†Senior Researcher, Aerospace Research and Development Directorate, 7-44-1 Jindaiji-Higashi-machi, Chofu-shi. Member AIAA.

‡Associate Professor, Institute of Space and Aeronautical Science, 3-1-1 Yoshinodai, Sagamihara-shi.

§Professor, Institute of Space and Aeronautical Science, 3-1-1 Yoshinodai, Sagamihara-shi.

I. Introduction

On 13 June 2010, the Hayabusa spacecraft returned to Earth and released its onboard capsule that could contain material from an asteroid [1]. The capsule then entered Earth's atmosphere with a velocity exceeding 12 km/s and successfully landed in the Australian outback. During the atmospheric reentry flight, the reentry capsule would experience significant aerodynamic heating due to its hypersonic entry speed. The flow temperature in the shock layer could become so high that air molecules could be nearly totally dissociated. For the thermal protection system (TPS) material, the Hayabusa capsule employed a conventional carbon fiber reinforced plastic (CFRP) ablator. The CFRP ablator consists of a base carbon fiber and a resin. When the virgin material of the ablator is heated, the resin decomposes and a pyrolysis gas is formed within the ablator. The pyrolysis gas then flows toward a wall surface and is released into the boundary layer. Besides, the ablating surface becomes nearly pure carbon due to the pyrolysis process. The solid carbon on the surface is oxidized or nitrided by the atomic oxygen or nitrogen in the boundary layer. As a result, the ablating surface recedes and the mass of the ablative TPS reduces during the entry flight. At present, a detailed inspection of the recovered ablative TPS for the Hayabusa reentry capsule has proceeded to quantify the amount of mass loss, surface shape changes, char layer depth, and so on [2]. In addition, radiation from the Hayabusa reentry capsule has been analyzed to estimate the surface temperature variation of the capsule along the reentry trajectory [2,3]. For design of the ablative TPS in future planetary missions, it is important to know that these flight data can be reproduced by the state-of-the-art numerical method and various physical models hitherto developed to predict the ablation phenomena.

As to the numerical methods of the thermal response of the ablator, one-dimensional computational codes were widely used in past analyses. Well-known examples are the Charring Materials Ablation code developed by Moyer and Rindal [4] and the Fully Implicit Ablation and Thermal Response program developed by Chen and Milos [5]. Other recent works include those by Ahn et al. [6], Suzuki et al. [7], and Amar et al. [8]. In these analysis methods, the thermal response of the ablator is calculated by giving a cold wall heat flux value as a code input. The boundary condition at the ablating surface, such as a blowing correction factor and a net heat flux transferred to the surface, is determined by a surface mass and energy balance [9].

Recently, a more detailed approach has been developed by integrating different computational tools [10]. In the method, the thermal response of the ablator is calculated by loosely coupling the shock layer computational fluid dynamics (CFD) code and the two-dimensional version of ablation code. This integrated analysis method was used to analyze the thermal responses of ablative test pieces that were put into arcjet hot airflow [11–13]. It was found from these studies that the time variations of the surface temperature of ablative test pieces were reproduced well by the present integrated analysis method. In addition, the temporal changes of the surface shape of the ablator test piece were successfully analyzed by the present method [13]. By using this integrated computational method, it seems possible now to evaluate the thermal response of the ablative TPS of the Hayabusa capsule more accurately.

The present study aims to analyze the experimental data obtained in the reentry flight of the Hayabusa capsule. For this purpose, the thermal response of the ablative TPS of the Hayabusa capsule is calculated along the most probable reentry trajectory by using the integrated analysis method.

Recently, several physical models have been proposed to predict the physical phenomena related to carbonaceous materials such as the CFRP ablator. One example is a probability model of the nitridation reaction that occurred at the graphite surface. In [14], the probability values of the nitridation reaction were estimated from the heating tests and the numerical model was developed based on the experimental data. In addition, a unique turbulence model was proposed by Park based on an assumption that the released pyrolysis gas could be already turbulent and could cause an increased heating rate [15]. Even though the turbulence model of Park has been successfully applied to explain the flight data obtained in Pioneer-Venus probes [16] and the Galileo probe [17], it is still meaningful to examine whether or not such models could reproduce the flight data obtained in the Hayabusa reentry mission. For these reasons, the calculations are performed by introducing these models into the integrated analysis method. The effects of the nitridation reaction and the transition to turbulence are examined through the comparison between the calculated results and the flight data.

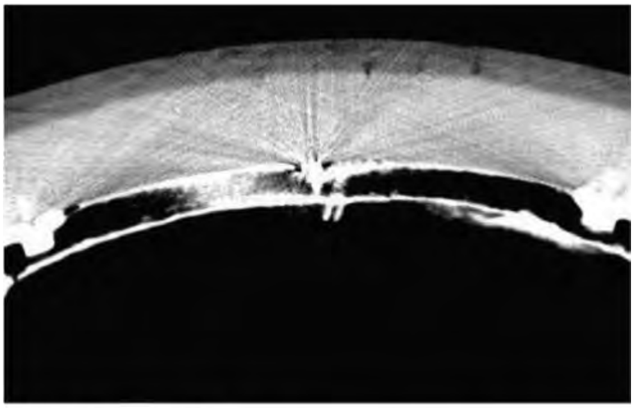
II. Postflight Analysis of Recovered Ablative TPS

To understand the flight environment of the Hayabusa capsule, the postflight analysis of the recovered ablative TPS is extremely important because the Hayabusa capsule itself did not take any measuring instruments on board. Although some of the details can be found in [2,3], the analysis procedure is briefly explained as follows.

The surface shape of the ablative TPS after the reentry flight was measured by using a three-dimensional laser scanner (Konica Minolta Range 7). The obtained results were compared with an initial shape to quantify the surface shape changes of the ablative TPS from before and after the reentry flight. It was found from the study that the recessed distance was about 0.3 mm at the stagnation region, whereas a slight thermal expansion was observed around the downstream region. It is believed that the phenolic resin inside the ablator expanded thermally under high temperature during the reentry flight.

An x-ray computed tomography (CT) method was also used to generate two-dimensional images inside the ablative TPS of the Hayabusa capsule. Typical examples of x-ray CT images are shown in Fig. 1. As shown in this figure, the difference between the char layer and the virgin layer can be recognized by the color difference. The density distribution inside the ablative TPS was then obtained by correlating a linear adsorption coefficient so measured with the density value of the ablative TPS. From the study, the thickness of the virgin layer was found to be larger than 16 mm for the stagnation region and 18 mm for the downstream region, respectively. Those measured results are compared with the calculated ones, and the results will be shown later.

Radiation emission from the Hayabusa reentry capsule was measured by the Japan Aerospace Exploration Agency (JAXA)/NASA joint airborne fireball observation team [2] and the JAXA ground observation team [3]. The temporal variation of the surface temperature of the Hayabusa reentry capsule was estimated based on the measured gray-body spectra. The quick reports can be found in [2,3], and the results are also compared with the calculated results in this study.



a) Near stagnation point

b) Near downstream region

Fig. 1 X-ray CT images inside the ablative TPS of the Hayabusa capsule.

III. Numerical Method

In the course of the simulation, the amount of mass loss of the ablative TPS of the Hayabusa reentry capsule is obtained by calculating the thermal response of the ablator through a coupled manner [10]. In the coupled method, the thermal response is analyzed by using different computational tools. Although some of the details are given in [10], a brief explanation of the procedure is made here for completeness.

A. Flowfield Analysis Around the Hayabusa Reentry Capsule

The shock layer CFD code developed earlier [18] for air and carbonaceous species is used for the computation of the dissociating, partially ionizing, and ablating flowfield over the Hayabusa reentry capsule. The flowfield over the Hayabusa reentry capsule is assumed to be in thermochemical nonequilibrium for an axisymmetric viscous flow. To calculate the nonequilibrium thermochemistry in the flowfield, Park et al.'s two-temperature model is used with 21 chemical species (N, O, N_2, O_2, NO, N^+, O^+, N_2^+, O_2^+, NO^+, C, C_2, CN, CO, C_3, C^+, H, H_2, C_2H, H^+, and e^-) and 36 chemical reactions [19]. The nitridation, oxidation, and sublimation reactions at the ablating surface are modeled in the calculation. These models will be explained later. The equations are discretized by the cell-centered finite volume scheme using the Advection Upstream Splitting Method-DV numerical flux [20] and MUSCL approach for attaining higher order spatial accuracy. Solutions are obtained by integrating the equations in time to steady state using the LU-SGS algorithm. It should be noted that the flowfield over the Hayabusa reentry capsule is calculated by accounting for the shape change of the capsule due to surface recession of the ablative TPS via the coupled calculation. This point will be explained later.

The net heat flux transferred to the wall surface is composed of the energy transport by convection and radiation and is given as follows:

$$q_{\text{net}} = q_t + q_{tv} + q_h + q_r - q_{rw} = \left(\kappa\frac{\partial T}{\partial n}\right) + \left(\kappa_v\frac{\partial T_v}{\partial n}\right) + \left(\sum_s \rho D_s\frac{\partial c_s}{\partial n}H_s\right) + q_r - \varepsilon\sigma T^4 \quad (1)$$

The radiative heat flux at the wall surface q_r is computed from the obtained flowfield data using the tangent-slab approximation. The emission and absorption coefficients are first calculated using a multiband model. In the present multiband model, the absorption coefficients are evaluated at 2294 wavelength points for air and at 7610 wavelength points for carbonaceous and hydrogenic species. They are constructed for the wavelength region from 750 to 15,000 Å. Other details can be found in [18].

B. Turbulence Model

It remains unknown whether the boundary layer around the Hayabusa capsule could be laminar or turbulent during the entry flight. According to Park et al., the released pyrolysis gas could be already turbulent and could cause an increased heating rate [15]. If this hypothesis is correct, some evidence (e.g., unexpected deep char layer, large amount of surface recession, and so on) might be observed in the recovered ablative TPS. For this case, those alterations in the ablative TPS cannot be predicted without taking account of such an early transition of the boundary layer. To discuss this possible higher heating due to the earlier transition of the boundary layer, the calculation using the injection-induced turbulence model [15] of Park is also carried out in this study. This turbulence model is based on the conventional mixing-length theory for characterizing the eddy viscosity of the released pyrolysis gas at the stagnation point. Because the concept of the injection-induced turbulence and the implementation of the model are described in detail in [16,18,21], the explanations are omitted here.

C. Thermal Response Analysis of Ablative TPS

The super charring materials ablation 2 (SCMA2) code developed earlier [10] is used to calculate the thermal response of the ablative TPS for the Hayabusa reentry capsule. In the SCMA2 code, two conservation equations for solid density and total energy are solved two dimensionally. The conservation equations are discretized by the cell-centered finite volume scheme. A pyrolysis rate source term is included in the right-hand side of the solid density conservation equation. The pyrolysis rate was determined by the least-squares method to reproduce the thermograms of the thermogravimetry data obtained in a different series of the experiment. The thermochemical properties, such as density, thermal conductivity, and specific heat of the ablator, are taken from [22]. According to [1,2], the Hayabusa capsule employed a tilted layup of carbon cloth with respect to the surface at the downstream region. The angle of cloth layers was 22.5 deg. For this reason, the effective thermal conductivity at the downstream region is believed to be different from that at the stagnation region. Such an effect is also included in the present study by taking account of a thermal conductivity tensor.

The thermal response analysis needs two boundary conditions: heat flux and mass flux values of chemical species produced by surface reactions. The heat flux and mass flux values at each position along the wall surface must be known as a function of time. The iteration method between the SCMA2 code and the shock layer CFD code over the Hayabusa reentry capsule is used to obtain the converged heat fluxes and mass flux values. The details of the method are shown later.

D. Mass Loss Due to Nitridation, Oxidation, and Sublimation Reactions

In the present study, the following three types of chemical reactions are considered to occur at the ablator surface: 1) nitridation by atomic nitrogen that produces CN, 2) oxidation by atomic oxygen that produces CO, and 3) sublimation that produces C_3. The mass flux of CN due to nitridation and that of CO due to oxidation are given as follows:

$$J_{\text{CN}} = \frac{M_{\text{CN}}}{M_N}\rho_N k \quad (2)$$

$$J_{\text{CO}} = \frac{M_{\text{CO}}}{M_O}\rho_O k \quad (3)$$

The symbol k denotes surface reaction velocity and is given by

$$k = \frac{\alpha}{4}\sqrt{\frac{8RT}{\pi M_s}} \quad (4)$$

where α and T denote the reaction probability and the surface temperature, respectively. The reaction probability of the oxidation reaction is taken from [23] and is expressed as

$$\alpha_{\text{oxi}} = 0.63\ \exp\left(-\frac{1160}{T}\right) \quad (5)$$

As for the probability of the nitridation reaction, the numerical model was proposed based on the experimental results [14] and is given by

$$\alpha_{\text{nit}} = 8.441 \times 10^{-3}\ \exp\left(-\frac{2322}{T}\right) \quad (6)$$

Mass flux of C_3 due to the sublimation reaction is given by the Hertz–Knudsen–Langmuir equation [24].

The reaction probability in Eq. (5) was usually measured by using smooth, solid carbon materials. However, the surface of ablative materials is most likely rough and has deep crevices. When deep crevices develop, an atomic species emerging from the inside crevice collides several times with another part of the wall inside the crevice before it leaves the crevice. As a result, the process occurring over the ablating surface with roughness accelerates the oxidation and nitridation. In a limiting case, the reaction probabilities could be taken to be unity [25] as follows:

$$\alpha_{\text{oxi}} = 1 \quad (7)$$

$$\alpha_{\text{nit}} = 1 \quad (8)$$

In the present study, some combinations of these probability models for oxidation and nitridation will be tested for the purpose of comparison. On the contrary, it is well known that surface roughness generally causes an earlier transition to turbulence. However, such an effect of the surface roughness on the transition to turbulence is beyond the scope of the present work and such a task will be left for the future.

The surface catalysis of carbonaceous materials such as the CFRP ablator is presently unknown. In the past, experimental studies were made to examine primary products in the attack of graphite by atomic oxygen [26–28]. According to these experiments, CO was the main product with a small amount of CO_2. As for the atomic nitrogen, the dissociated nitrogen flow around the ablating graphite was analyzed by using the present integrated analysis method, and the results are compared with the measured ones [29]. It was found from the study that the temporal variations of surface temperature during the heating test and the surface recession distributions of the graphite test piece could be reasonably reproduced by the calculations, assuming that the graphite surface was noncatalytic to atomic nitrogen. From these experimental and numerical studies, it is believed that catalytic recombination reaction by atomic species hardly occurs at the carbonaceous materials surface. For this reason, the ablator surface is assumed to be noncatalytic to atomic nitrogen and atomic oxygen for all the computed cases in the present study.

The mass loss rate at each grid point is obtained from the CFD solutions as

$$r = \frac{M_C}{M_{\mathrm{CN}}}\frac{J_{\mathrm{CN}}}{\rho_c} + \frac{M_C}{M_{\mathrm{CO}}}\frac{J_{\mathrm{CO}}}{\rho_c} + \frac{M_C}{M_{C_3}}\frac{J_{C_3}}{\rho_c} \tag{9}$$

The amount of surface recession at a certain time point is obtained by integrating the mass loss rate with time and is given as follows:

$$S = \int_0^{\mathrm{Time}} r\,\mathrm{d}t \tag{10}$$

E. Computational Meshes and Flight Trajectory of Hayabusa Reentry Capsule

The Hayabusa capsule has an axisymmetric blunt configuration with a nose radius of 0.2 m. Typical examples of computational meshes for both the flowfield and the ablative TPS are shown in Fig. 2. The number of grid points used is 51×71 for the flowfield and 51×101 for the ablator, respectively. The solution was also obtained on a finer computational mesh system with four times the number of grid points for both the flowfield (101×141) and the ablator (101×201), as compared with the standard mesh system given in Fig. 2. The surface temperature variation calculated by using the finer mesh system was compared with that obtained by using the standard mesh system. It was found that the difference of the surface temperature at the peak heating point is less than 1%. For this reason, the results will be presented by using the standard mesh.

The freestream condition is applied to the outer boundary of the computational mesh for the flowfield. The boundary condition between the ablative TPS and the flowfield around the Hayabusa capsule is calculated using a coupled manner. The details of the coupling method will be introduced in the beginning of Sec. IV. A radiative cooling effect is also considered at the inner boundary of the computational mesh for the ablative TPS.

The flight trajectory used in this study is given by [3] and the representative parameters are shown in Fig. 3. The time evolution of the flight environment is obtained by using the atmospheric model based on the global reference atmospheric model (GRAM-99) and

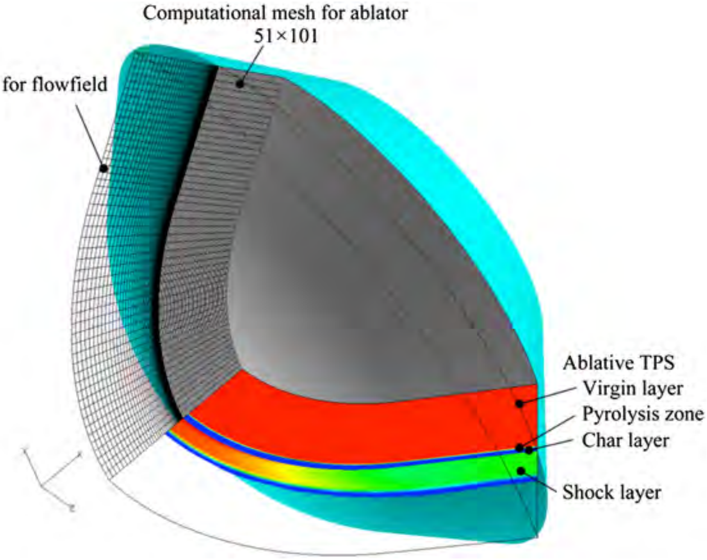


Fig. 2 Computational meshes for flowfield calculation and thermal response analysis for Hayabusa reentry capsule. Typical example of converged solution at $t = 70$ s from 200 km is also shown.

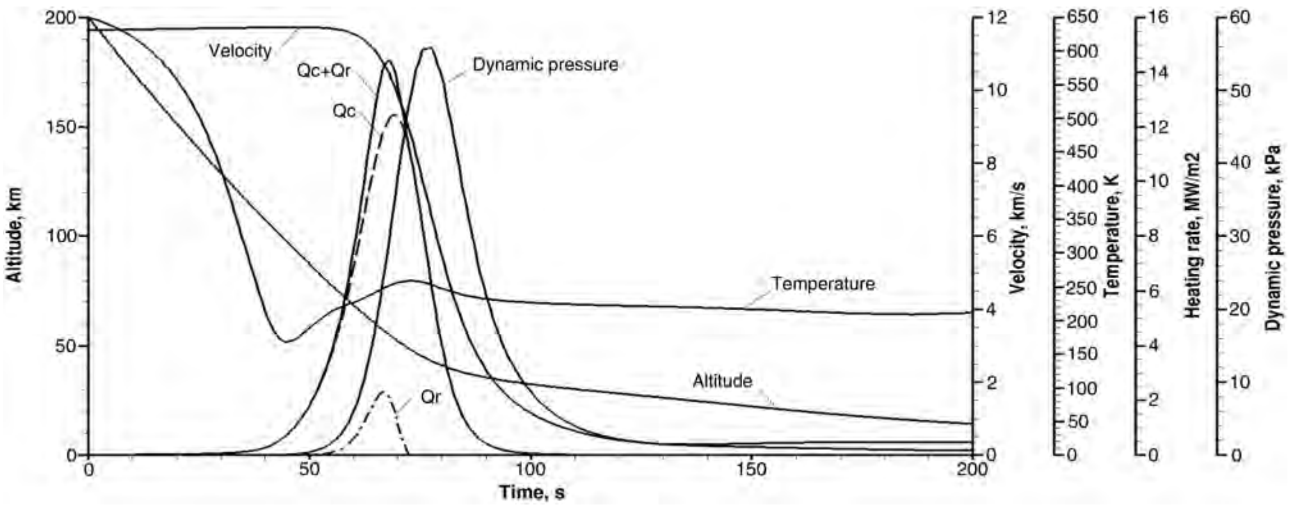


Fig. 3 Most probable flight trajectory of Hayabusa reentry capsule [3].

Table 1 Freestream conditions at CFD calculation points

UTC, hh:mm:ss	Flight time, s	Altitude, km	Velocity, km/s	Density, kg/m^3	Knudsen number	Temperature, K
13:52:06	55	77.75	11.695	2.408×10^{-5}	6.71×10^{-2}	213.7
13:52:11	60	68.59	11.546	9.065×10^{-5}	1.80×10^{-3}	226.8
13:52:16	65	59.94	11.061	2.897×10^{-4}	5.71×10^{-4}	242.3
13:52:21	70	52.16	9.868	7.785×10^{-4}	2.15×10^{-4}	256.4
13:52:26	75	45.74	7.807	1.785×10^{-3}	9.37×10^{-5}	258.1
13:52:31	80	40.99	5.422	3.516×10^{-3}	4.72×10^{-5}	247.7

predicted nominal entry conditions. In this figure, the flight time is counted from 13 June 2010, 13:51:11 Coordinated Universal Time (UTC), at which time the Hayabusa capsule passed through 200 km altitude. In Fig. 3, the cold wall heating rate and the radiative heating rate are roughly estimated using the Detra–Kemp–Riddell equation [30] and Tauber et al.'s empirical formulas [31,32], respectively. As shown in this figure, the total heating rate reaches its maximum at 70 s of flight time. The flowfield calculation around the Hayabusa reentry capsule is made every 5 s from 55 to 80 s of flight time. The freestream flow conditions at these points are summarized in Table 1.

The boundary-layer Reynolds number of the Hayabusa capsule is estimated to be 1.7×10^{-4} at the peak heating point. Generally speaking, boundary transitions hardly occur at a low Reynolds number on the order of 10^{-4}. However, several experimental studies on the laminar-to-turbulent transitions predicted the transition at $\mathrm{Re} = 3 \times 10^{-4}$ on a blunt body with surface mass injection [33,34]. It implies that the boundary layer transition might occur during the entry flight of the Hayabusa capsule. In this study, the evidence of the boundary-layer transition will be examined through the comparison between the calculations and the flight data.

Not shown in Fig. 3, the angle of attack of the Hayabusa capsule was found to be less than 2 deg from 50 to 100 s of flight time. It is believed that axisymmetric analysis made in this study would be valid under such a low angle-of-attack condition.

F. Calculation Conditions

In the present study, the thermal response of the ablative TPS of the Hayabusa capsule is calculated by using several physical models related to the ablation phenomena. Calculation conditions used in this study are summarized in Table 2. Four cases are considered for the purpose of comparison: In case 1, the conventional surface reaction set consisting of oxidation and sublimation reaction is considered. It is well known that this conventional model was also used to calculate the heating environments over the Stardust entry capsule [35]. The effect of the nitridation reaction is considered in case 2. In case 3, the effect of the enhanced probability value due to surface roughness is considered, both for atomic oxygen and atomic nitrogen. The effect of turbulence due to the pyrolysis gas is included in case 4. A typical example of the calculated solution for case 1 is shown in Fig. 2. In this figure, the pressure contour of the flowfield and the density contour inside the ablative TPS at 70 s of flight time are shown.

IV. Results and Discussions

In the course of the present calculation, an initial CFD solution is first obtained at the first trajectory point of 55 s of flight time. Ablation and surface reactions are assumed to be absent. The constant wall temperature is arbitrarily chosen as 3000 K. The boundary conditions for the SCMA2 code, such as the convective and radiative heat flux distributions along the wall surface, are obtained from this initial solution. The heat flux values are assumed to be zero before 40 s (altitude is above 100 km and density is below 1.0×10^{-7} kg/m^3) for convenience and are interpolated by a quadratic function up to the first trajectory point at 55 s. By using the interpolated heat flux values from 40 to 55 s of flight time, the SCMA2 code solves the thermal response of the ablative TPS along the reentry trajectory and gives the wall boundary conditions, such as wall temperature and the pyrolysis gas injection rate, from 40 to 55 s. Once the wall boundary condition at 55 s is specified, the new CFD solution with ablation and surface reactions is calculated at the same trajectory point. The converged solution then gives the new boundary condition for the SCMA2 code at 55 s, and the thermal response of the ablative TPS is again calculated along the trajectory. This iteration procedure is repeated until the difference of the heat flux values at each grid point becomes less than 0.1% compared with those at a previous iteration. In the present study, it is found that three or four iterations are needed to obtain convergence. Once the converged solution is obtained at 55 s, the calculation is carried out at the next trajectory point of 60 s. As before, the initial CFD solution is obtained without ablation and surface reactions at 60 s. The heat flux values are linearly interpolated from 55 to 60 s, which are fed into the SCMA2 code. The solution from the SCMA2 code then gives the new boundary condition for the CFD code. The converged and consistent solution with ablation at 60 s is finally obtained iteratively. The converged solutions at 65, 70, 75, and 80 s of flight time are similarly obtained by using the converged solution at the previous trajectory point.

Figure 4 shows the calculated heat flux values at the stagnation point from 55 to 80 s for the case 1. The results are presented for the initial solution and the converged solution between the CFD and the SCMA2 codes. For the converged solution, the heat flux values defined by Eq. (1) are shown in the figure. As can be seen, the convective heat flux reaches its maximum value of 5.3 MW/m^2 at 70 s, whereas the maximum radiative heat flux value of about 1 MW/m^2 is obtained at 65 s of flight time.

Figure 5 shows the temporal variations of surface temperature at the stagnation point. As shown in this figure, the temperature begins to increase at 40 s. A maximum temperature value of about 3200 K is then obtained at 70 s of flight time for cases 1, 2, and 4. After that time, the surface temperature decreases due to the radiative cooling effect. As shown in this figure, there are no significant differences in surface temperature between cases 1 and 2. The effect of the nitridation reaction on surface temperature is believed to be small because the rate of nitridation given by Eq. (6) is smaller than the rate of oxidation given by Eq. (5) by a factor of 100. The results for the turbulent case (case 4) will be discussed later.

Table 2 Calculation conditions

Case	Surface reaction probability			Boundary layer	Notes
	Oxidation	Nitridation	Sublimation		
1	Eq. (5)	0	Included	Laminar	Conventional surface reaction set
2	Eq. (5)	Eq. (6)	Included	Laminar	Case 1 + nitridation
3	Eq. (7)	Eq. (8)	Included	Laminar	Fully rough surface for O and N
4	Eq. (5)	0	Included	Turbulent	Case 1 + turbulence due to pyrolysis gas

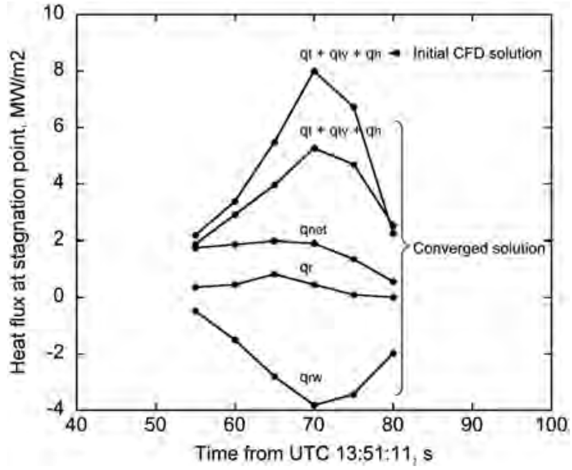


Fig. 4 Temporal variations of heat flux value at the stagnation point for case 1.

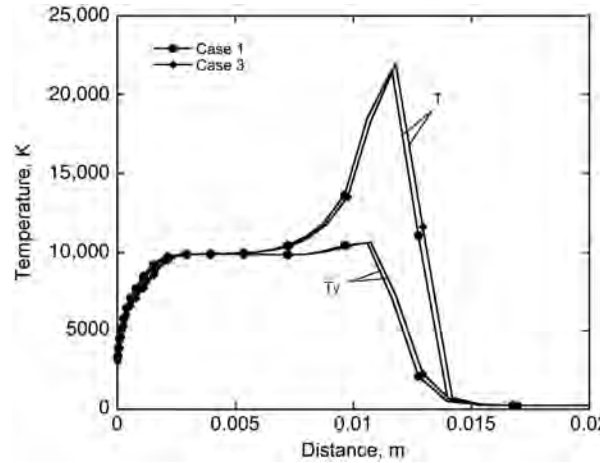


Fig. 7 Comparison of temperature along stagnation streamline between case 1 and case 3 at 70 s from 200 km altitude.

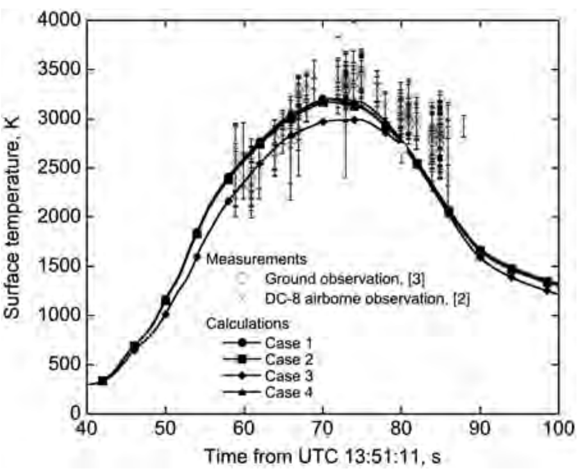


Fig. 5 Comparison of surface temperature variations at the stagnation point between the calculation and the measurements.

From Fig. 5, one can see that the computed peak temperature for case 3 becomes lower by about 200 K compared with other cases. This lower temperature is explained as follows: The summation of mass flux values produced by the oxidation J_{CO}, nitridation J_{CN}, and sublimation J_{C3} are plotted against time in Fig. 6. As shown in Fig. 6,

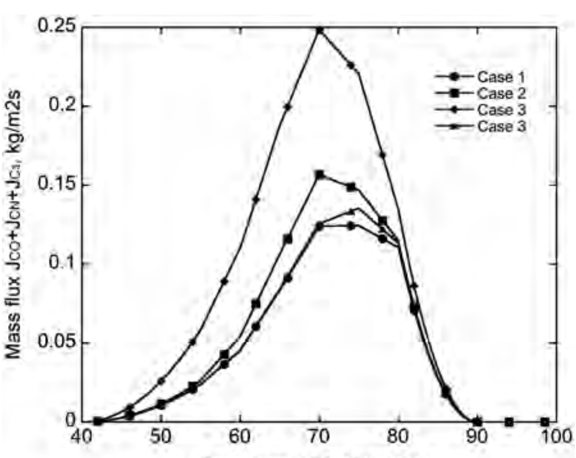


Fig. 6 Comparison of mass flux of chemical species produced at the wall surface.

the mass flux value for case 3 becomes two times larger than that for case 1. This trend is because the reaction probability values of oxidation and nitridation reactions for case 3 are larger than those for case 1. For this reason, it is considered that a convective blockage effect for case 3 becomes significant compared with that for any other cases. Figure 7 shows the temperature distributions along the stagnation streamline at 70 s of flight time for cases 1 and 3. Because of the large convective blockage effect, one can see that the temperature gradient near the wall surface for case 3 becomes smaller than that for case 1. This large convective blockage effect, in turn, affects the heat flux transferred to the wall. The distributions of heat flux components at 70 s are shown in Figs. 8a–8d. As shown in Fig. 8a, the heat flux value due to the translational–rotational temperature gradient q_t for case 3 becomes lower than that for case 1. As a result, the heat flux transferred to the wall by convection $(q_t + q_{tv} + q_h)$ for case 3 becomes smaller than that for case 1, as shown in Fig. 8d. For this reason, the surface temperature for case 3 becomes lower than that for any other cases.

Figure 5 also shows the surface temperature data estimated from the radiation emission obtained by both the DC-8 airborne observation and the ground observation [2,3]. As shown in this figure, the temperature value estimated from the DC-8 airborne observation data is about 2500 K at 60 s of flight time, and the maximum value of 3300 K is obtained at 70 s. After that, the temperature decreases down to about 2700 K at 85 s, although the temperature values are characterized by very strong scattering after the peak heating point. Error bars indicating ± 1 standard error are also presented in Fig. 5. On the other hand, the surface temperature is estimated to be about 2750 ± 150 K at 80 s from the ground observation result [3]. By comparing these estimated surface temperatures with the present calculated results, it is found that the calculation for cases 1, 2, and 4 duplicates well the temperature data obtained by the airborne observation, although the calculated peak temperatures slightly underestimate the measurement at 70 s of flight time. After that, the calculated temperature agrees well with the temperature data obtained by the ground observation, whereas it is slightly lower than the temperature given by the airborne observation. It is considered that reasonable agreement is obtained between the calculated temperature and temperature values deduced from the optical data for the cases 1, 2, and 4.

From Fig. 8d, one can see that the heat flux distribution for case 4 is substantially different from that for any other cases. This trend is because the heating rate in the downstream region becomes large due to the earlier transition of the boundary layer. As a result, the heat flux value in the downstream region becomes comparable with that at the stagnation point at 70 s of flight time. It should be noted that the result obtained in the present study is also seen in the works by Takahashi and Sawada [16] and Suzuki et al. [18].

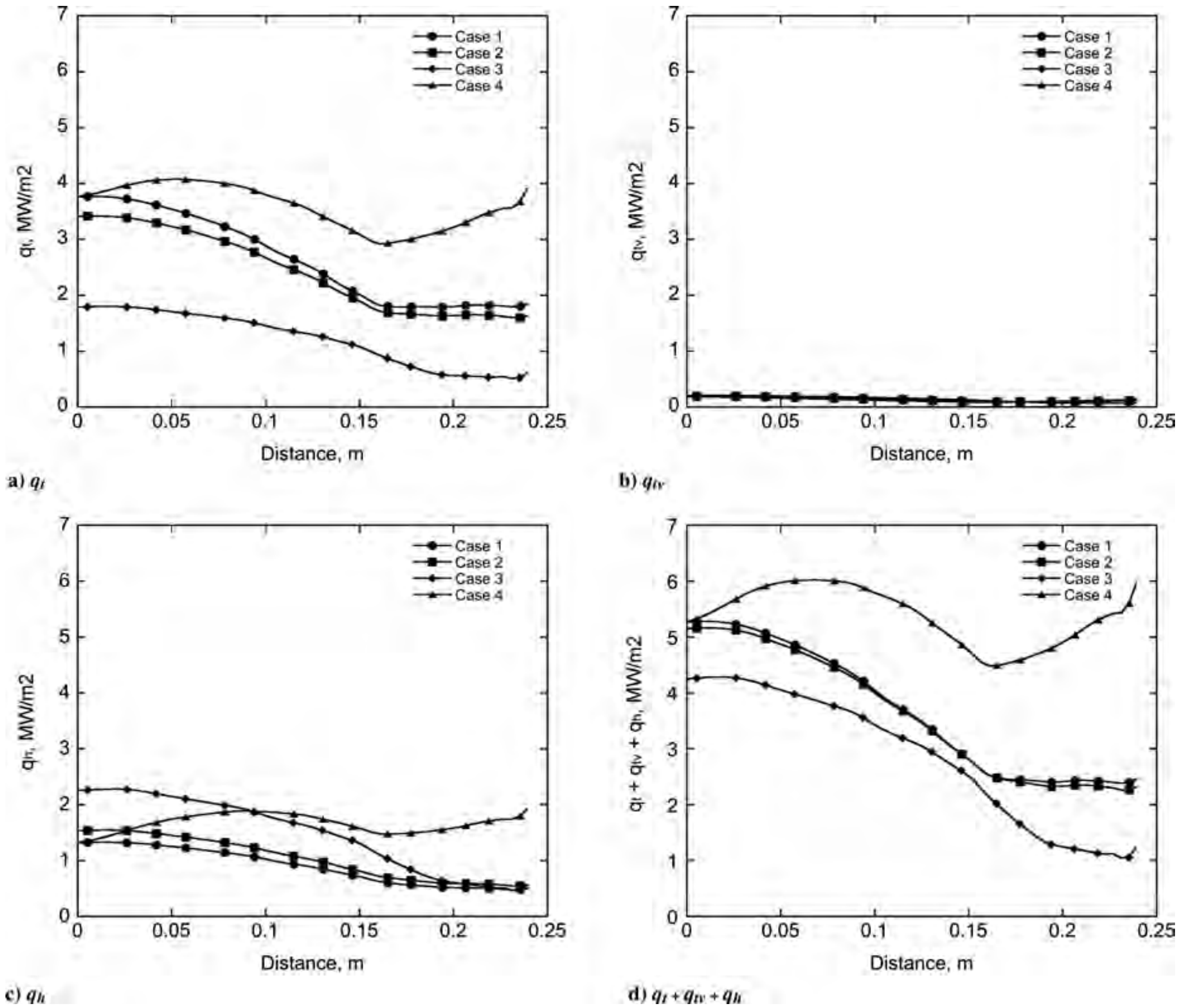


Fig. 8 Heat flux distributions along wall surface at 70 s of flight time.

The surface temperature distributions along the wall surface at each CFD calculation point are shown in Figs. 9a–9f. For cases 1, 2, and 3, the surface temperature reaches maximum at the stagnation point and then gradually decreases to the downstream region at all calculation points. For case 4, in contrast, the surface temperature becomes larger at about 0.07 m downstream from the stagnation point and then gradually decreases toward the juncture point of 0.16 m, but slightly increases again at the frustum edge. This qualitative trend becomes significant particularly at 70 and 75 s from 200 km altitude. The reason for this high temperature appearing in the downstream region is that the heating rate in the downstream region becomes large due to the turbulent heating, as was shown in Fig. 8d. As a result, the surface temperature in the downstream region becomes comparable with that at the stagnation point at all calculation points.

Figure 10 shows the time variation of surface recession at the stagnation point. As shown in Fig. 10, the surface recession is about 1.05 mm for case 1. The amount of surface recession for case 2 is about 20% larger than that for case 1. The reason for this difference is because the amount of mass loss of the ablative TPS increases due to nitridation reaction, as shown in Fig. 6. The surface recession further increases and reaches up to 2.06 mm, if the effect of the enhanced probability value due to surface roughness is considered both for atomic oxygen and atomic nitrogen. The calculated recession profiles along the wall surface after the reentry flight are shown in Fig. 11. From Fig. 11, one can see that the surface recession for case 4 becomes larger than that of any other cases in the downstream region. As a result, the amount of surface recession becomes larger than 1 mm along the entire wall surface. This trend is because of the higher temperature due to the turbulent heating, as was shown in Fig. 9.

According to the preliminary shape measurement using the three-dimensional laser scanner, the surface recession was about 0.3 mm at the stagnation region, whereas surface recession was not observed around the downstream region. The result is also presented in Fig. 11. A maximum error in the shape measurement is estimated to be less than $\pm 10\%$. Based on these results, it is considered that the present calculation overestimates the flight data not only at the stagnation region but also in the downstream region. The reason for this difference in the surface recession is currently unknown. One possible reason is that the ablative TPS of the Hayabusa capsule expanded thermally due to the high temperature during the reentry flight. In that case, the difference in the surface recession could be improved by taking account of thermal expansion inside the ablator. This point needs to be confirmed in the future.

The density distributions inside the ablative TPS at the stagnation point ($s = 0$ m) and the downstream region ($s = 0.2$ m) are shown in Figs. 12a and 12b, respectively. From Fig. 12a, one can see that the calculated density distribution for case 1 duplicates well that for cases 2 and 4. This trend is because there were no significant differences in the temporal variations of surface temperature and the amount of surface recession between those cases, as was shown in Figs. 9 and 10. Note that the thickness of the virgin layer for case 3 becomes close to that for case 1, although the amount of surface recession for case 3 is about two times larger than that for case 1, as shown in Fig. 10. This trend is believed to occur because the surface temperature for case 3 is substantially lower than that for case 1, as shown in Fig. 9. As a result, the temperature distribution inside the ablator for case 3 might

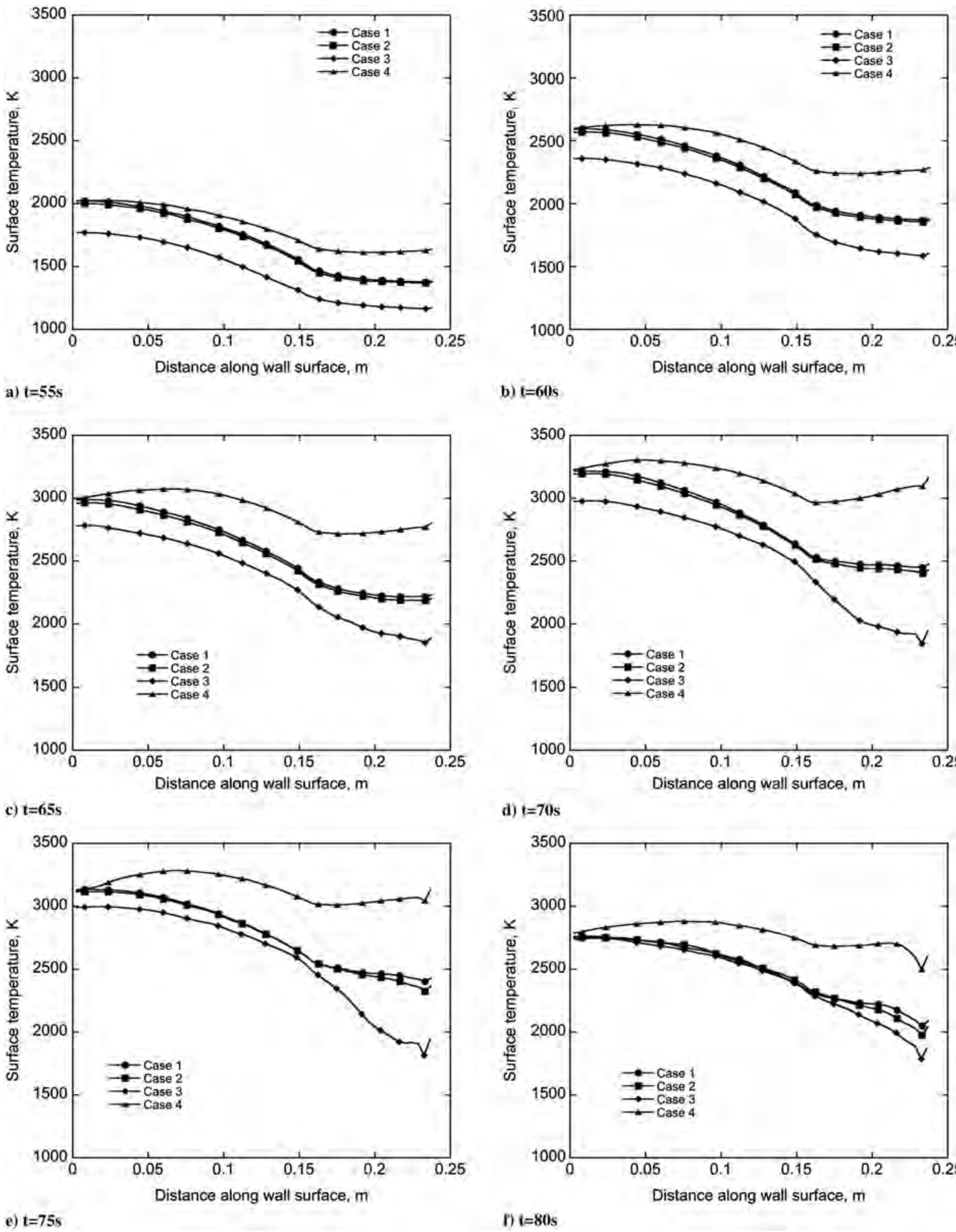


Fig. 9 Surface temperature distributions along wall surface at chosen calculation points.

become comparable with that for case 1. On the other hand, significant differences can be seen between the calculated results in Fig. 12b. As shown in this figure, the char depth for case 4 becomes larger than that for cases 1 and 2. In addition, the thickness of the virgin layer for case 4 becomes smaller than that for any other cases. This trend is because the surface temperature in the downstream region is significantly elevated by the effect of turbulence due to the pyrolysis gas. On the contrary, the thickness of the virgin layer for case 3 becomes larger than that for case 1 because of the lower temperature due to the large convective blockage effect, as shown in Fig. 8.

Based on the x-ray CT image inside the ablative TPS of the Hayabusa capsule shown in Fig. 1, the thickness distributions of the char and virgin layers were evaluated along the ablator surface. It was found that the char depth was about 5.5 mm and the thickness of the virgin layer was about 16.5 mm at the stagnation point ($s = 0$ m). In contrast, the thickness values of the char and virgin layers were 4.0 and 18.5 mm for the downstream region ($s = 0.2$ m). Those values

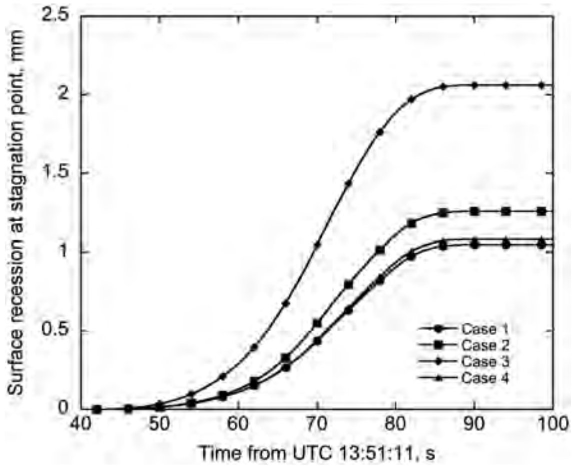


Fig. 10 Time variations of surface recession at the stagnation point.

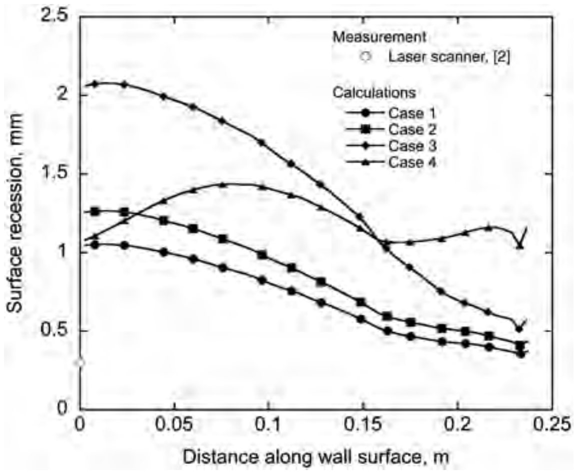
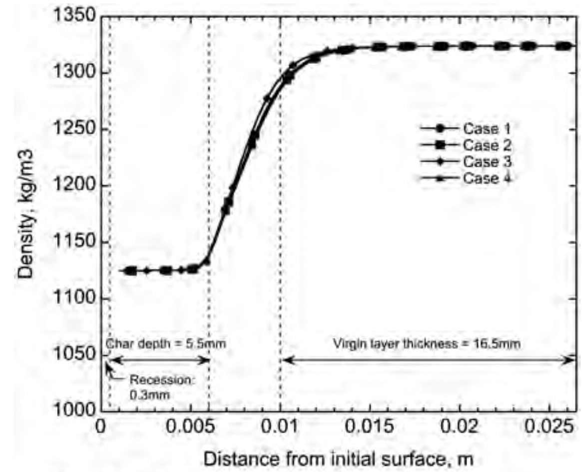


Fig. 11 Computed surface recession along wall surface at 300 s of flight time.

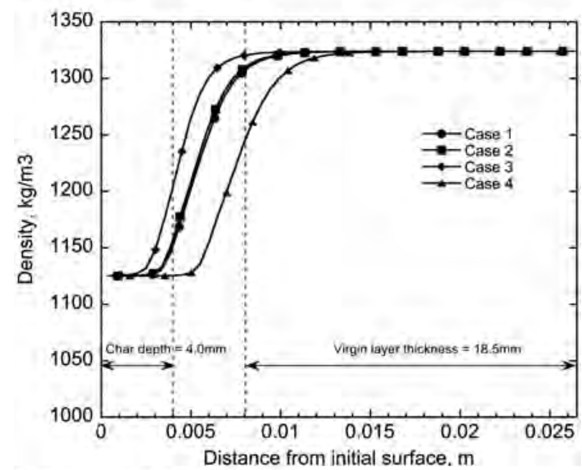
are also presented in Figs. 12a and 12b, respectively. The experimental uncertainty is less than ± 0.5 mm.

Careful attention should be paid to compare the thickness of the virgin and char layers between the flight data and the present calculations, because the present calculation did not take account of the thermal expansion of the ablative TPS and overestimated the surface recession along the entire wall surface. However, the difference in surface recession between the flight data and the calculation is not more than 1 mm even at the stagnation point for case 1. In addition, the amount of thermal expansion could be roughly estimated to be less than 0.5 mm along the wall surface under the linear thermal expansion coefficient of $5.0 \times 10^{-6}\ 1/\mathrm{K}$, the change in temperature of 3000 K, and the typical thickness of the ablator of 30 mm. It is considered that the thickness of the virgin and char layers would be large enough compared with these experimental uncertainties. For this reason, it is believed that the direct comparison between the calculation and the flight data would be still meaningful.

By comparing the flight data and the calculated results in Figs. 12a and 12b, it is found that the density distributions calculated for cases 1 and 2 reproduce the flight data reasonably, both at the stagnation point and the downstream region. Especially for the thickness of the char layer in the downstream region, the calculation overestimates the flight data when the effect of turbulence due to the pyrolysis gas is considered, whereas it underestimates the flight data when the effect of surface roughness of the ablative TPS is taken into account. Based on these results, it is considered that the thickness values of the virgin and char layers could be reproduced well by taking account of the conventional surface reaction set and nitridation reaction, whereas it seems unlikely that the early transition to turbulence could occur during the reentry flight of the Hayabusa capsule.



a) Stagnation point ($s = 0m$)



b) Downstream region ($s = 0.2m$)

Fig. 12 Comparison of char depth and thickness of virgin layer between the calculations and the flight data.

V. Conclusions

A trajectory-based thermal response analysis of the Hayabusa ablative thermal protection system (TPS) is performed by loosely coupling the axisymmetric thermochemical nonequilibrium computational fluid dynamics code and the super charring materials ablation 2 code that gives the thermal response of the ablative TPS. The radiative heat flux is computed by using the tangent-slab approximation with the multiband model. Several physical models related to the ablation phenomena, such as a nitridation reaction, surface roughness, and earlier transition of boundary layer due to the ablation product gas, are tested by introducing the models into the present analysis method.

The surface recession calculated by taking account of the nitridation reaction becomes 20% larger than that calculated without nitridation, whereas there are no significant differences in surface temperature between cases with and without nitridation reaction. The surface recession calculated by considering the surface roughness of the ablative TPS becomes two times larger than that calculated without the surface roughness effect. On the contrary, the surface temperature calculated with surface roughness is substantially lower than that calculated without the surface roughness due to the large convective blockage effect. By taking account of the effect of earlier transition of the boundary layer due to the pyrolysis gas injection, the surface temperature in the downstream region is significantly elevated, resulting in a large amount of surface recession and a deep pyrolysis front.

The present calculated results are compared with the flight data obtained at the present time. Reasonable agreement is obtained between the calculated surface temperature and the temperature values estimated from optical data obtained by DC-8 airborne and ground observations. It is also found that the calculated surface recession overestimates the measurement along the wall surface by a factor of three, even if the nitridation reaction is not taken into account in this study. On the contrary, the density distributions inside the ablator calculated by taking account of the nitration reaction reproduce the flight data reasonably, both at the stagnation point and the downstream region. As for the thickness of the char layer in the downstream region, the calculation overestimates the flight data when the effect of turbulence due to the pyrolysis gas is considered, whereas it underestimates the flight data when the effect of surface roughness of the ablative TPS is taken into account.

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References

[1] Yamada, T., Ishii, N., and Inatani, Y., "Post-Flight Analysis of the Hayabusa Sample Return Capsule," *28th International Symposium on Space Technologies*, Paper 2011-def-03, 2011.

[2] Yamada, T., Ishii, N., and Inatani, Y., "Post-Flight Analysis of the Recovered Heatshield of Hayabusa Sample Return Capsule," *Journal of Spacecraft and Rockets*, AIAA Paper 2011-3322, 2011.

[3] Fujita, K., Suzuki, T., Matsuyama, S., and Abe, S., "Numerical Reconstruction of HAYABUSA Sample Return Capsule Flight Environments," *Journal of Spacecraft and Rockets*, AIAA Paper 2011-3477, 2011.

[4] Moyer, C. B., and Rindal, R. A., "An Analysis of the Coupled Chemically Reacting Boundary Layer and Charring Ablator, Part II: Finite Different Solution for the In-Depth Response of Charring Materials Considering Surface Chemical and Energy Balances," NASA CR-1061, June 1968.

[5] Chen, Y.-K., and Milos, F. S., "Ablation and Thermal Response Program for Spacecraft Heatshield Analysis," *Journal of Spacecraft and Rockets*, Vol. 36, No. 3, 1999, pp. 475–483.
doi:10.2514/2.3469

[6] Ahn, H.-K., Park, C., and Sawada, K., "Response of Heatshield Material at Stagnation Point of Pioneer-Venus Probes," *Journal of Thermophysics and Heat Transfer*, Vol. 16, No. 3, 2002, pp. 432–439.
doi:10.2514/2.6697

[7] Suzuki, T., Sawada, K., Yamada, T., and Inatani, Y., "Experimental and Numerical Study of Pyrolysis Gas Pressure in Ablating Test Piece," *Journal of Thermophysics and Heat Transfer*, Vol. 19, No. 3, 2005, pp. 266–272.
doi:10.2514/1.12211

[8] Amar, A., Blackwell, B., and Edwards, J., "Development and Verification of a One-Dimensional Ablation Code Including Pyrolysis Gas Flow," *Journal of Thermophysics and Heat Transfer*, Vol. 23, No. 1, 2009, pp. 59–71.
doi:10.2514/1.36882

[9] Potts, R. L., "Hybrid Integral/Quasi-Steady Solution of Charring Ablation," AIAA Paper 90-1677, June 1990.

[10] Suzuki, T., Sakai, T., and Yamada, T., "Calculation of Thermal Response of Ablator Under Arcjet Flow Condition," *Journal of Thermophysics and Heat Transfer*, Vol. 21, No. 2, 2007, pp. 257–266.
doi:10.2514/1.25499

[11] Suzuki, T., Fujita, K., Sakai, T., Okuyama, K., Kato, S., and Nishio, S., "Prediction Accuracy of Thermal Response of Ablator Under Arcjet Flow Conditions," AIAA Paper 2010-1174, 2010.

[12] Suzuki, T., Fujita, K., Sakai, T., Okuyama, K., Kato, S., and Nishio, S., "Evaluation of Prediction Accuracy of Thermal Response of Ablator Under Arcjet Flow Conditions," AIAA Paper 2010-4787, 2010.

[13] Suzuki, T., Fujita, K., Sakai, T., Okuyama, K., Kato, S., and Nishio, S., "Thermal Response Analysis of Low-Density CFRP Ablator," *Transactions of the Japan Society for Aeronautical and Space Sciences, Aerospace Technology Japan*, Vol. 10, 2012, pp. Pe_21–Pe_30.
doi:10.2322/tastj.10.Pe_21

[14] Suzuki, T., Fujita, K., and Sakai, T., "Graphite Nitridation in Lower Surface Temperature Regime," *Journal of Thermophysics and Heat Transfer*, Vol. 24, No. 1, 2010, pp. 212–215.
doi:10.2514/1.43265

[15] Park, C., "Injection-Induced Turbulence in Stagnation Point Boundary Layer," *AIAA Journal*, Vol. 22, No. 2, 1984, pp. 219–225.
doi:10.2514/3.8371

[16] Takahashi, M., and Sawada, K., "Simulation of Entry Flight Flowfield over Four Probe Vehicles in Pioneer-Venus Mission," AIAA Paper 2002-0909, 2002.

[17] Matsuyama, S., Ohnishi, N., Sasoh, A., and Sawada, K., "Numerical Simulation of Galileo Probe Entry Flowfield with Radiation and Ablation," *Journal of Thermophysics and Heat Transfer*, Vol. 19, No. 1, 2005, pp. 28–35.
doi:10.2514/1.10264

[18] Suzuki, T., Furudate, M., and Sawada, K., "Unified Calculation of Hypersonic Flowfield for a Reentry Vehicle," *Journal of Thermophysics and Heat Transfer*, Vol. 16, No. 1, 2002, pp. 94–100.
doi:10.2514/2.6656

[19] Park, C., Jaffe, R. L., and Partridge, H., "Chemical-Kinetic Parameters of Hyperbolic Earth Entry," *Journal of Thermophysics and Heat Transfer*, Vol. 15, No. 1, 2001, pp. 76–90.
doi:10.2514/2.6582

[20] Wada, Y., and Liu, M. S., "A Flux Splitting Scheme with High Resolution and Robustness for Discontinuities," AIAA Paper 94-0083, Jan. 1994.

[21] Izawa, Y., and Sawada, K., "Calculation of Surface Heat Transfer for Sphere with Wall Injection," *Journal of Thermophysics and Heat Transfer*, Vol. 14, No. 2, 2000, pp. 230–236.
doi:10.2514/2.6513

[22] Yamada, T., Inatani, Y., and Hirai, K., "Thermal Responses of Ablator for Reentry Capsules with Superorbital Velocity," Inst. of Space and Astronautical Science Rept. SP-17, March 2003, pp. 201–215.

[23] Park, C., "Effect of Atomic Oxygen in Graphite Ablation," *AIAA Journal*, Vol. 14, No. 11, 1976, pp. 1640–1642.
doi:10.2514/3.7267

[24] Blottner, F. G., "Prediction of Electron Density in the Boundary Layer on Entry Vehicles with Ablation," NASA SP-252, 1970, pp. 219–240.

[25] Park, C., *Nonequilibrium Hypersonic Aerothermodynamics*, Wiley, New York, 1989, pp. 350–353.

[26] Allendorf, H. D., and Rosner, D. E., "Primary Products in the Attack of Graphite by Atomic Oxygen and Diatomic Oxygen about 1100K," *Carbon*, Vol. 7, No. 4, 1969, pp. 515–518.
doi:10.1016/0008-6223(69)90086-4

[27] Rosner, D. E., and Allendorf, H. D., "High Temperature Oxidation of Carbon by Atomic Oxygen," *Carbon*, Vol. 3, No. 2, 1965, pp. 153–156.
doi:10.1016/0008-6223(65)90042-4

[28] Blackwood, J. D., and McTaggart, F. K., "Oxidation of Carbon with Atomic Oxygen," *Australian Journal of Chemistry*, Vol. 12, No. 2, 1959, pp. 114–121.
doi:10.1071/CH9590114

[29] Suzuki, T., Fujita, K., and Sakai, T., "Experimental Study of Graphite Ablation in Nitrogen Flow, Part II: Further Numerical Analysis," *Journal of Thermophysics and Heat Transfer*, Vol. 24, No. 3, 2010, pp. 589–597.
doi:10.2514/1.43264

[30] Detra, R. W., Kemp, N. H., and Riddell, F. R., "Addendum to Heat Transfer to Satellite Vehicles Re-Entering the Atmosphere," *Jet Propulsion*, Vol. 27, No. 12, 1957, pp. 1256–1257.

[31] Tauber, M. E., Bowles, J. V., and Yang, L., "Use of Atmospheric Braking During Mars Missions," *Journal of Spacecraft and Rockets*, Vol. 27, No. 5, 1990, pp. 514–521.
doi:10.2514/3.26174

[32] Tauber, M. E., and Sutton, K., "Stagnation-Point Radiative Heating Relations for Earth and Mars Entries," *Journal of Spacecraft and Rockets*, Vol. 28, No. 1, 1991, pp. 40–42.
doi:10.2514/3.26206

[33] Kaattari, G. E., "Effect of Mass Addition on Blunt-Body Boundary-Layer Transition and Heat Transfer," NASA TP-1139, Jan. 1978.

[34] Demetriades, A., Laderman, A. J., Von Seggern, L., Hopkins, A. T., and Donaldson, J. C., "Effect of Mass Addition on the Boundary Layer of a Hemisphere at Mach 6," *Journal of Spacecraft and Rockets*, Vol. 13, No. 8, 1976, pp. 508–509.
doi:10.2514/3.27924

[35] Olynick, D., Chen, Y.-K., and Tauber, M. E., "Aerothermodynamics of the Stardust Sample Return Capsule," *Journal of Spacecraft and Rockets*, Vol. 36, No. 3, 1999, pp. 442–462.
doi:10.2514/2.3466

G. Palmer
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